Lesson Planning

Complete Lesson Plan Book

Class: Class 4

Subject: Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Food and Nutrients

Learning Objectives

• Name the main nutrients in food.

• Tell which foods give energy.

• Choose a balanced diet.

Key Concepts

• Carbohydrates give us energy: rice, bread.

• Proteins build our body: eggs, meat, beans.

• Vitamins protect us from illness: fruits.

• Water and milk keep us strong.

Teaching Methodology (Step by Step)

1. Show food pictures; pupils name nutrients.
2. Sort foods into energy, body-building, protective.
3. Discuss: what did you eat today?
4. Make a balanced meal plate.
5. Taste test: identify food groups.

Classroom Activities

• Balanced plate collage.

• My food diary for three days.

Assessment Questions

1. Name three nutrients.
2. Which food gives energy?
3. Why do we need vitamins?
4. Name two protein foods.

Homework

• Write what you ate today; label the nutrients.

Time Allocation

4 periods of 35 minutes each.

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Unit 2: How Plants Grow

Learning Objectives

• Name what plants need to grow.

• Describe germination simply.

• Care for a growing plant.

Key Concepts

• Plants need water, sunlight, air and soil.

• A seed sprouts into a seedling.

• Roots grow down; shoots grow up.

• Plants make their own food.

Teaching Methodology (Step by Step)

1. Soak beans; observe sprouting daily.
2. Plant seeds in cups; measure growth.
3. Discuss: what happens without sunlight?
4. Draw the life cycle of a plant.
5. Water the class plants on a rota.

Classroom Activities

• Seed germination experiment.

• Plant growth chart with measurements.

Assessment Questions

1. Name four things plants need.
2. What comes out of the seed first?
3. Which way do roots grow?
4. How do plants make food?

Homework

• Grow a bean at home; record growth for a week.

Time Allocation

4 periods of 35 minutes each.

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Unit 3: Animals: Babies and Food

Learning Objectives

• Name babies of common animals.

• Tell what different animals eat.

• Sort animals: plant-eaters and meat-eaters.

Key Concepts

• A baby cat is a kitten; a dog's is a puppy.

• Cows and goats eat plants.

• Lions and eagles eat meat.

• Some animals eat both.

Teaching Methodology (Step by Step)

1. Match babies with parent animals.
2. Sort animal cards: plant-eaters vs meat-eaters.
3. Tell: how do mothers care for babies?
4. Discuss pets' food needs.
5. Draw an animal family.

Classroom Activities

• Baby animal matching game.

• Visit a farm (or picture tour).

Assessment Questions

1. What is a baby cow called?
2. Name two plant-eating animals.
3. What do lions eat?
4. Name the baby of a hen.

Homework

• Draw three animals with their babies; label them.

Time Allocation

3 periods of 35 minutes each.

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Unit 4: Solids, Liquids and Gases

Learning Objectives

• Identify solids, liquids and gases.

• Tell properties of each state.

• Give daily examples of each.

Key Concepts

• Solids keep their shape: book, stone.

• Liquids flow and take the shape of containers.

• Gases spread everywhere: air.

• Water can be solid ice, liquid, or steam.

Teaching Methodology (Step by Step)

1. Sort objects into three groups.
2. Pour water into different containers.
3. Feel air in a balloon.
4. Freeze water; observe ice melting.
5. Discuss: steam from hot tea.

Classroom Activities

• State-sorting relay.

• Make ice cubes and observe melting.

Assessment Questions

1. Name two solids.
2. What shape does a liquid take?
3. Name one gas.
4. Is ice a solid or liquid?

Homework

• List five solids, five liquids and two gases at home.

Time Allocation

4 periods of 35 minutes each.

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Unit 5: Push and Pull (Force)

Learning Objectives

• Show push and pull with examples.

• Tell how force changes motion.

• Understand heavy things need more force.

Key Concepts

• Push moves things away; pull brings closer.

• Force can start, stop or change motion.

• Heavy objects need bigger pushes.

• Friction slows moving things.

Teaching Methodology (Step by Step)

1. Push and pull classroom objects.
2. Roll balls: gentle vs strong push.
3. Tug of war game.
4. Slide objects on rough vs smooth surfaces.
5. Discuss: where do we use force daily?

Classroom Activities

• Toy car ramp races.

• Push-pull sorting cards.

Assessment Questions

1. Give an example of a push.
2. Give an example of a pull.
3. What slows a rolling ball?
4. Which needs more force: a book or a desk?

Homework

• List five pushes and five pulls at home.

Time Allocation

3 periods of 35 minutes each.

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Unit 6: Light and Shadows

Learning Objectives

- Tell that light helps us see.
- Make shadows with objects.
- Explain day and night simply.

Key Concepts

- The sun is our main source of light.
- Shadows form when light is blocked.
- Shadows change with the sun's position.
- Day and night come from Earth's rotation.

Teaching Methodology (Step by Step)

1. Darken the room; use a torch.
2. Make hand shadows on the wall.
3. Go outside: observe shadows at noon.
4. Discuss: why is night dark?
5. Draw day and night scenes.

Classroom Activities

- Shadow puppets show.
- Track your shadow at three times of day.

Assessment Questions

1. What is our main source of light?
2. How is a shadow formed?
3. Why is it dark at night?
4. Name two sources of light at home.

Homework

- Draw an object and its shadow; label the light source.

Time Allocation

3 periods of 35 minutes each.

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Unit 7: Land and Water on Earth

Learning Objectives

• Name the main landforms.

• Tell the difference between seas and rivers.

• Locate land and water on a globe.

Key Concepts

• Mountains are high lands; valleys are low.

• Rivers flow; seas are salty and vast.

• Most of Earth is covered with water.

• Pakistan has mountains, plains and deserts.

Teaching Methodology (Step by Step)

1. Show landforms with pictures.
2. Build mountains and valleys with clay.
3. Find blue (water) and green/brown (land) on a globe.
4. Discuss: where does our drinking water come from?
5. Draw land and water scenes.

Classroom Activities

• Clay landform models.

• Globe exploration: find oceans.

Assessment Questions

1. Name two landforms.
2. What is the difference between a river and a sea?
3. What covers most of Earth?
4. Name one landform found in Pakistan.

Homework

• Draw a scene with one landform and one water body.

Time Allocation

3 periods of 35 minutes each.

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